

Saadbin Khan

📍 Blacksburg, VA | 🌐 **Personal Website** | ✉️ saadbin@vt.edu | 🔗 LinkedIn | 🎓 Google Scholar | 🏠 ResearchGate

HIGHLIGHTS

- 9+ years of experience in CFD simulations and multiscale heat and mass transport modeling using OpenFOAM, Python, Fortran, C++, MATLAB, ANSYS Fluent, and SolidWorks, spanning nanoscale Stokes regime to atmospheric turbulence.
- 7+ years of experience in data-driven modeling, numerical engineering analysis, multivariate analysis, non-linear regression, data processing, visualization, and interpretation.
- 5+ years of interdisciplinary research experience at the interface of computational mechanics and biological sciences.
- 4+ years of experience in SolidWorks CAD; custom meshing with *snappyHexMesh*; geometry processing and repair with Meshmixer; and data visualization and post-processing using ParaView.
- 1 year of experience with particle-based rarefied gas simulations using Direct Simulation Monte Carlo (DSMC) method.

EDUCATION

Ph.D. in Engineering Mechanics

2021 - 2026 (exp.)

Virginia Tech (CGPA: 3.89)

Blacksburg, VA

Dissertation: The role of hydrodynamic slip and fine tracheal geometry in insect respiratory flow.

Notable courses: Finite Element Method, Fluid & Solid Mechanics, Continuum Mechanics, Intermediate Dynamics.

M.S. in Mechanical and Aerospace Engineering

2017 - 2019

Oklahoma State University

Stillwater, OK

Thesis: Near wall structure and deviation from equilibrium in wavy channel turbulence. 🔗

Notable courses: CFD, Turbulence, Numerical Engineering Analysis, Applied Computing, Linear & Non-linear Programming.

B.Sc. in Mechanical Engineering

2011 - 2016

Bangladesh University of Engineering and Technology

Dhaka, Bangladesh

Thesis: Mixed convection & MHD effect analysis in a lid-driven cavity with sinusoidally curved bottom wall using CNT-water nanofluid in a porous medium. 🔗

Notable courses: Thermodynamics, Heat Transfer, Mechanical Drawing, Machine Design, IC Engines, Fluidics.

TECHNICAL STRENGTHS

Programming:	Python, Fortran, MATLAB, C++, C, Mathematica
Design & Simulation:	SolidWorks, OpenFOAM, ANSYS Fluent, COMSOL, DSMC, OpenLB, Open MPI
Database & Post-processing:	Tecplot, Paraview, Microsoft Excel, Tecplot, Tracker, ImageJ

NOTABLE PROJECTS

- ◇ **The role of hydrodynamic slip and fine tracheal geometry in insect respiratory flow** [Portfolio: 🔗] 2022 - 2026
 - Funder: NSF | PI: Anne Staples | Tools: OpenFOAM, C++, Python, SolidWorks, DSMC, OpenLB, Paraview
 - Outcome: Demonstrated that slip and fine wall structures enhance tracheal flow beyond classical no-slip predictions; consequently developed a microscale-physics informed modified Murray's law applicable to microfluidic devices.
- ◇ **Waviness quantification of collagen fibers from simple harmonic generation (SHG) images** [Portfolio: 🔗] 2022
 - Funder: BIOTRANS | PI: Raffaella De Vita | Tools: Python, CT-FIRE
 - Outcome: Quantified collagen fiber straightness and orientation from 20 SHG images using CT-FIRE; evaluated software limitations, demonstrating increased failure rates for highly wavy fibers and non-uniform width or opacity.
- ◇ **DNS of wavy channel flow to understand the near-wall turbulence structure** [Portfolio: 🔗] 2017 - 2019
 - Funder: NASA Oklahoma EPSCoR | PI: Balaji Jayaraman | Tools: Fortran, Python, Open MPI, ANSYS Fluent
 - Outcome: Simulated 3D turbulent flow at $Re \approx 2800$ over sinusoidal roughness using MPI parallelization with HPC, showing significantly stronger Reynolds stress distributions and turbulence structures relative to flat-wall channel flow.
- ◇ **Improved atmospheric plume source inversion model for remote leak detection** [Portfolio: 🔗] 2018
 - Funder: Baker Hughes General Electric | PI: Jamey Jacob | Tools: Python
 - Outcome: Achieved up to 99.2% source localization accuracy using only 1% concentration measurement data in a structured 2D domain.
- ◇ **Mixed convection heat transfer in nanofluids with MHD effects in a porous medium** [Portfolio: 🔗] 2016
 - PI: Mohammad Arif Hasan Mamun | Tools: COMSOL, Tecplot
 - Outcome: Identified optimal heat transfer performance at 15% waviness to cavity width ratio and 9% CNT-nanoparticle concentration, with consistent increase in average Nusselt number across operating conditions

RELEVANT EXPERIENCE

- ◇ **Research Assistant, Laboratory for Fluid Dynamics in Nature** 2021 - Present
Virginia Tech Blacksburg, VA
 - Simulated 3D CFD models of insect respiratory transport to quantify the effects of slip and fine tracheal structures.
 - Developed a microscale-physics informed modified Murray's law that applies to biological transport networks.
 - 1 *Nature Reviews Physics* paper (March '25 cover article and featured in Virginia Tech newsletter), 1 *Nature Physics* paper in final phase, 1 other journal paper in review, 9 talks in conferences & symposiums.
- ◇ **Instructor of Record, Department of Mechanical Engineering** 2025
Virginia Tech Blacksburg, VA
 - Courses taught: Thermodynamics (ME-2134).
- ◇ **Teaching Assistant** 2017 - 2019; 2021 - Present
Oklahoma State University; Virginia Tech Stillwater, OK; Blacksburg, VA
 - Courses assisted: Thermodynamics, Fluid Mechanics–Control Volumes, Dynamics, Engineering Analysis of Physiological Systems, Engineering Analysis & Methods.
- ◇ **Research Assistant, Biological Transport Program** 2021 - 2022
Virginia Tech Blacksburg, VA
 - Characterized uterosacral ligaments using straightness and orientation metrics using SHG image stacks in CT-FIRE, evaluated software performance, and identified failure modes for highly wavy and non-uniform fibers.
 - Operationalized a Leica RM2125RTS rotary microtome and established protocols for repeatable tissue sectioning.
- ◇ **Research Assistant, Computational & Flow Science Research Laboratory** 2017 - 2019
Oklahoma State University Stillwater, OK
 - Developed Fortran subroutines in the Incompact3d framework for non-flat surface flows; developed Python post-processing pipelines; performed data-driven analysis for LES-relevant turbulence modeling.
 - Implemented a Python-based forward atmospheric dispersion model using 3D unsteady advection–diffusion equations to improve upon traditional Gaussian dispersion models; developed a sparse-reconstruction-based plume source inversion model in Python for remote leak detection using very limited concentration data.
 - 2 journal papers, 1 symposium talk, 1 code partially utilized in BHGE's 'LUMEN' project.
- ◇ **Engineering Intern** 2016
Bangladesh Atomic Energy Commission Dhaka, Bangladesh
 - Inspected structural safety, control-rod operation, and cooling water regulation in a 3 MW nuclear research reactor.
- ◇ **Other Services**
 - **Journal Reviewer** for *Fluids*, MDPI; *Partial Differential Equations in Applied Mathematics*, Elsevier.
 - **Webmaster** for the leadership team in Graduate Engineering Mechanics Society, Virginia Tech, Blacksburg, VA, USA.
 - **Student Administrative Assistant** in Biological Transport program, Virginia Tech, Blacksburg, VA, USA.
 - **Public Outreach Manager** in Bangladeshi Students Association, Oklahoma State University, Stillwater, OK, USA

SELECTED PEER-REVIEWED PUBLICATIONS [FULL LIST: [Google Scholar](#) (280+ CITATIONS)]

- [Nature Physics (*In review*)] *S Khan*, AE Staples, “A tidal Murray's law framework inspired by insect respiration”.
- [Nature Reviews Physics 2025, Cover Article] *S Khan*, AE Staples, “Mechanisms of insect respiration”. 🔗
- [Nanoparticle-based polymer composites 2022] *S Khan*, MK Hossain, “Classification & properties of nanoparticles”. 🔗
- [AIP Advances 2020] B Jayaraman, *S Khan*, “Direct numerical simulation of turbulence over two-dimensional waves”. 🔗
- [Fluids 2019] *S Khan*, B Jayaraman, “Statistical structure and deviations from equilibrium in wavy channel turbulence”. 🔗
- [AIP Conference Proceedings 2016] MI Khan, KM Rabbi, *S Khan*, M Mamun, “Mixed convection analysis in lid-driven cavity with sinusoidally curved bottom wall using CNT-water nanofluid”. 🔗

AWARDS

- **Alice & Dan Pletta Scholarship**, Engineering Mechanics Graduate Program, Virginia Tech 2026
- **BIOTRANS Fellowship**, Interdisciplinary Graduate Education Programs, Virginia Tech 2021 - 2022
- **Champion**, 6th and 8th Bangladesh Mathematical Olympiad 2008 & 2010

(References available upon request)